

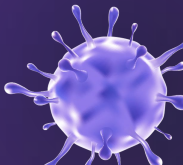


Covid 19 detection using CNN and ResNet

Introduction



- Chest X-rays (CXR) are widely used for detecting lung-related abnormalities due to their speed, affordability, and availability.
- Manual interpretation of chest X-rays is time-consuming and dependent on radiologist expertise.
- Early-stage infections often show subtle patterns that are difficult to distinguish visually.
- COVID-19 symptoms in X-rays can appear similar to other lung infections (e.g., pneumonia), leading to misclassification.
- There is a need for an automated, accurate, and fast detection system using deep learning to assist radiologists in diagnosis.

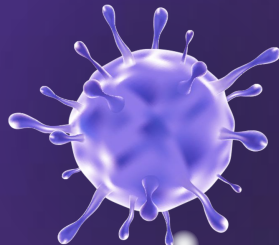


DATASET DESCRIPTION



Covid-19	3616
Normal	10192
Lung-opacity(n on covid lung infection)	6012
Viral Pneumonia	1345

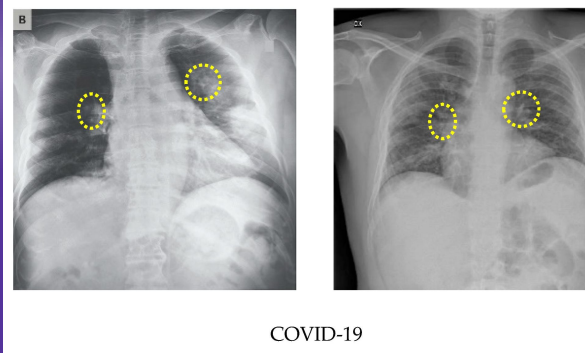
- The dataset consists of chest radiography images (X-rays) used for disease detection.
- Images are grouped into four distinct classes for classification tasks.
- Includes lung masks for selected images to support segmentation.
- Collected from publicly available medical image sources.



X-rays

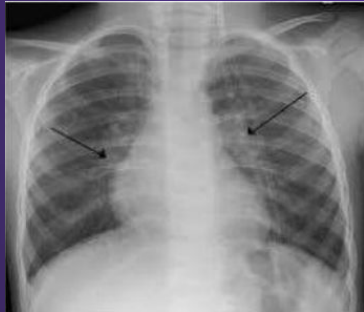


normal

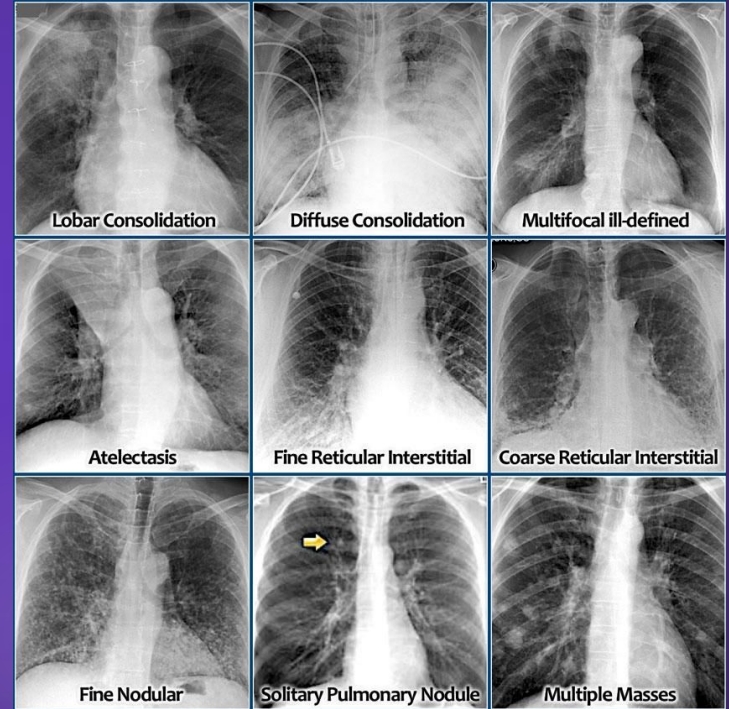


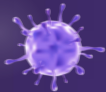
COVID-19

covid

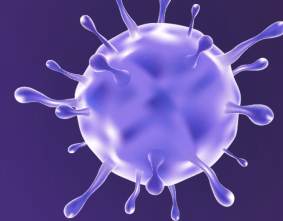


Viral pneumonia





WHY THIS DATASET? WHY DLH?



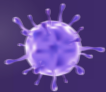
The COVID-19 Radiography Dataset is selected because it is highly suitable for building healthcare models.

1. Multi-class dataset
2. Large dataset size
3. Availability of labeled data
4. Supports multiple tasks
5. Publicly available and widely used

Why DLH?

1. High accuracy in image classification
2. Handles complex medical data
3. Reduces human effort
4. Scalability
5. Improves healthcare decision-making





Dataset Handling

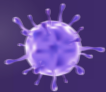
- Dataset loaded from Google Drive
- Extracted ZIP file into working directory
- Verified dataset structure
- Checked class distribution

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Data Splitting

- Train: 70%
- Validation: 15%
- Test: 15%
- Random shuffling applied
- Separate folders created





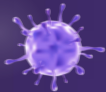
Data Preprocessing and Augmentation

- **Pixel Normalization**
Ensures all input data is on a **uniform scale** [0,1]
- **Pixel Value Rescaling**
Pixel values normalized using **rescale = 1/255** for faster and stable training
- **Rotation**
Random rotation applied up to **20°** to improve model robustness.
- **Zoom**
Random zoom applied with a range of **0.2** to capture scale variations.
- **Horizontal Flip**
Images are flipped horizontally to increase dataset diversity.



Convolutional Neural Network (CNN) for Medical Image Classification





Model Compilation

- Adam optimizer + Cross Entropy loss define learning and error calculation
- Accuracy metric used to evaluate model performance

Training Process

- Trained for 10 epochs
- Used validation data for monitoring
- Accuracy tracked during training

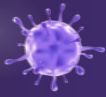
Accuracy Results

- Training Accuracy: ~80.99%
- Validation Accuracy: ~85.85%

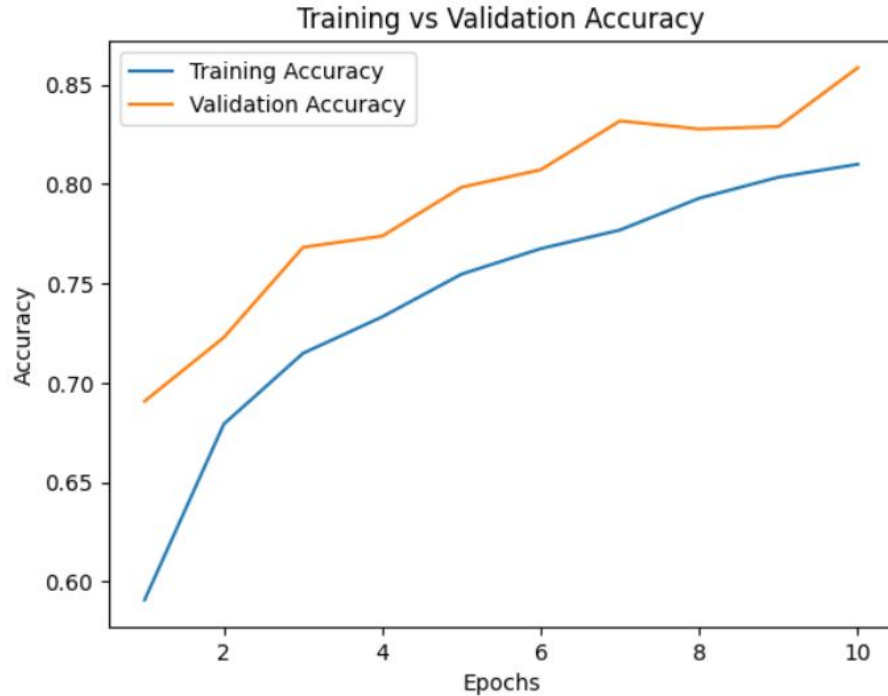
Testing Accuracy

- Testing Accuracy: ~86.24%
- Represents real-world model performance





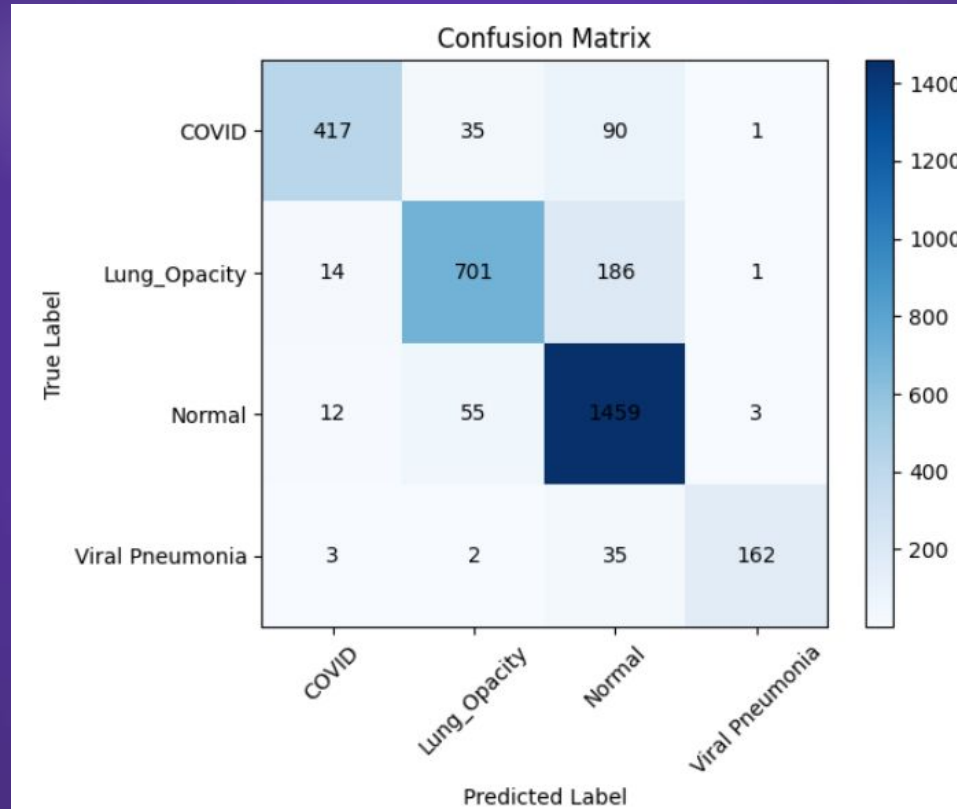
Accuracy Graph

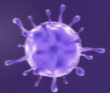


Loss Graph



Confusion Matrix





CNN Model –

Training Optimization (EarlyStopping)

- EarlyStopping used to prevent overfitting
- Monitors validation loss with patience of 3 epochs
- Automatically restores best model weights

Why Training Stopped Early



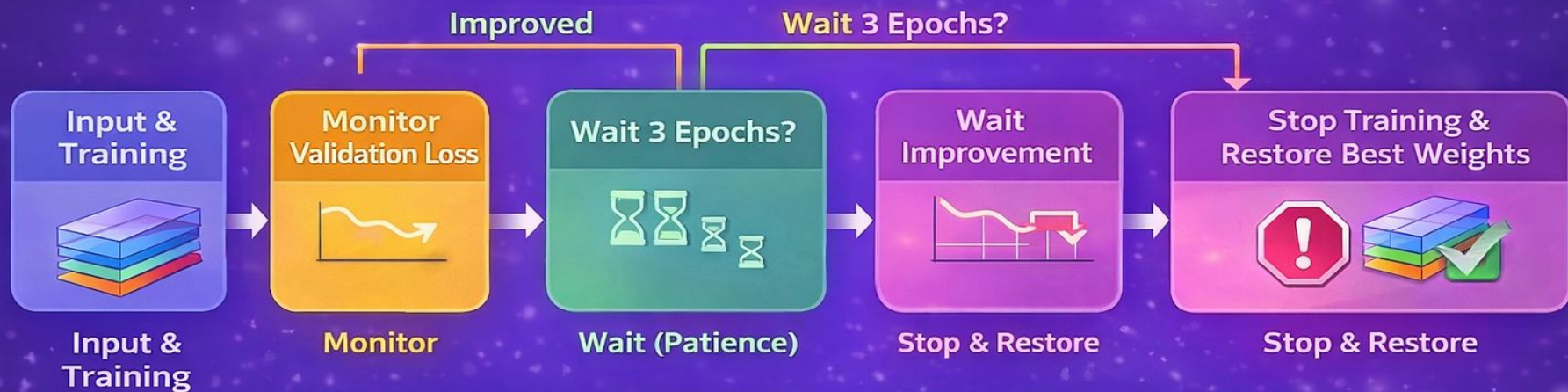
- Training stopped at optimal epoch (around Epoch 12)
- Validation loss plateaued (no further improvement)
- EarlyStopping triggered to avoid overfitting

Training Configuration

- Epochs: 20 with controlled training duration
- Batch size: 32 for efficient computation
- Adam optimizer with Categorical Cross Entropy loss

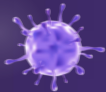


Training Optimization with EarlyStopping



■ Input & Training ■ Monitor ■ Wait (Patience) ■ Stop & Restore

- ✓ Monitors: Validation Loss
- Patience: 3 Epochs
- No Improvement: Stops Training
- Restores Best Model Weights



Accuracy Results

- Training Accuracy $\approx 80.997\%$
- Validation Accuracy $\approx 85.85\%$

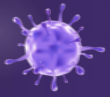
.



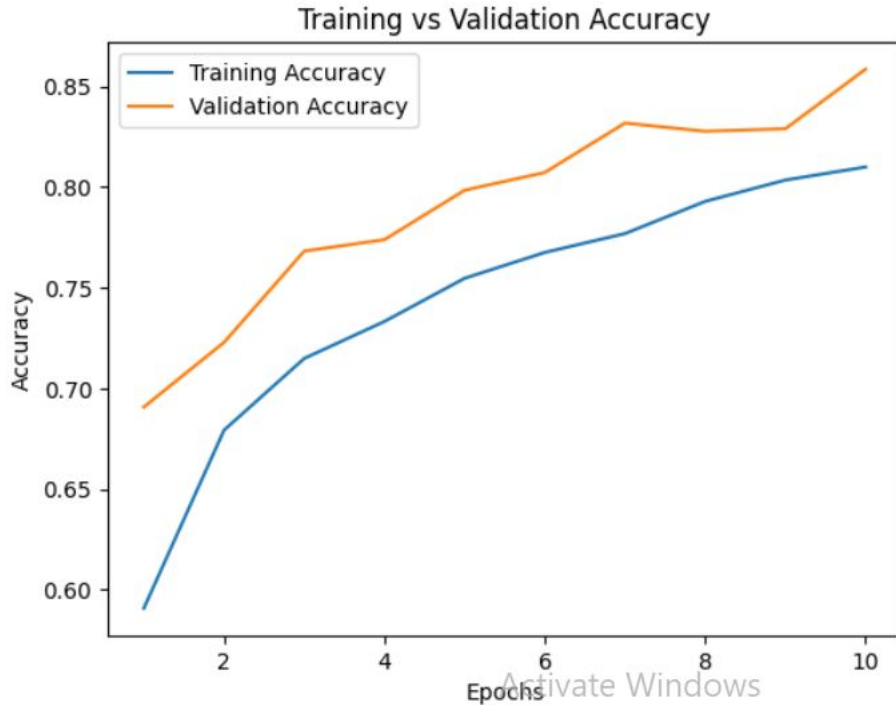
Testing Accuracy

- Testing Accuracy: $\sim 88.53\%$
- Indicates strong performance on unseen data





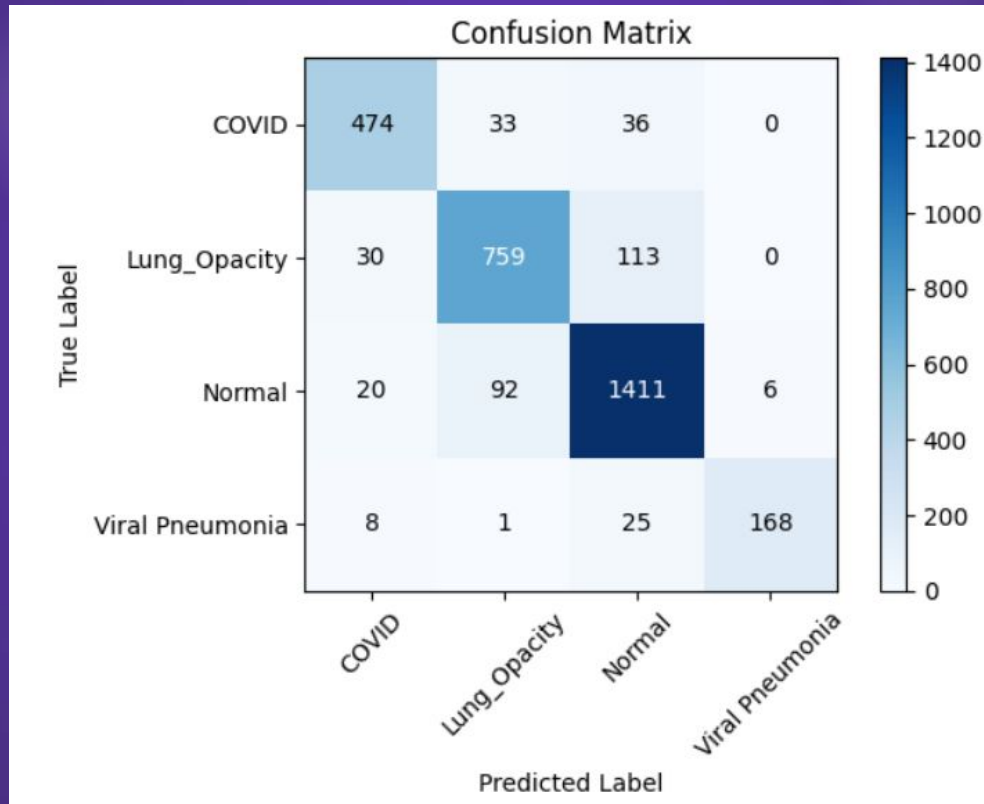
Accuracy Graph



Loss Graph



Confusion Matrix



Why Transfer Learning?

Using X-ray Images



Pretrained Knowledge

Leverages feature extraction capabilities learned from large-scale datasets like **ImageNet** (1M+ images).



Faster Convergence

Significantly reduces training time as lower layers already detect edges, textures, and shapes effectively.



Works with Limited Data

Achieves high performance even when medical imaging datasets are small, preventing **overfitting** common in deep scratch-built networks.



Domain Adaptation

Enables fine-tuning where specific layers are retrained to adapt general visual features to **chest X-ray** image characteristics.



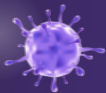


ResNet 50

BLOCK DIAGRAM :

- We used ResNet50, a deep Convolutional Neural Network (CNN) architecture.
- We used it as a feature extractor for X-Ray images classification.
- Base model + GAP + Dense(128,ReLU) + Dense(4,Softmax)





Model Performance Before **Fine-Tuning**



Model **compilation**

- Optimizer: **Adam**
- Loss Function: **Categorical Crossentropy**
- Metric: **Accuracy**

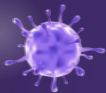
TRAINING- **PROCESS**

- Epochs: **10**
- Batch size: **32**
- **Training + Validation**



Accuracy **Results**

- Training Accuracy: **~88.12 %**
- Validation Accuracy: **~89.89 %**
- Test Accuracy : **~90.3 %**



Model Performance After Fine-Tuning



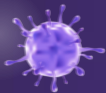
Fine Tuning Process

- Unfroze last 30 layers of ResNet50
- Reduced learning rate to $1e-5$
- Retrained model for 10 epochs



Accuracy Results

- Training Accuracy: **~93.07 %**
- Validation Accuracy: **~92.72 %**
- Test Accuracy : **~93.36 %**

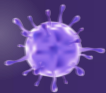


Comparison of Model Performance

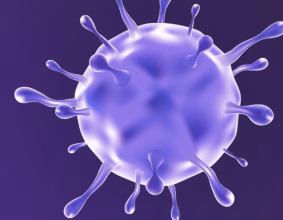
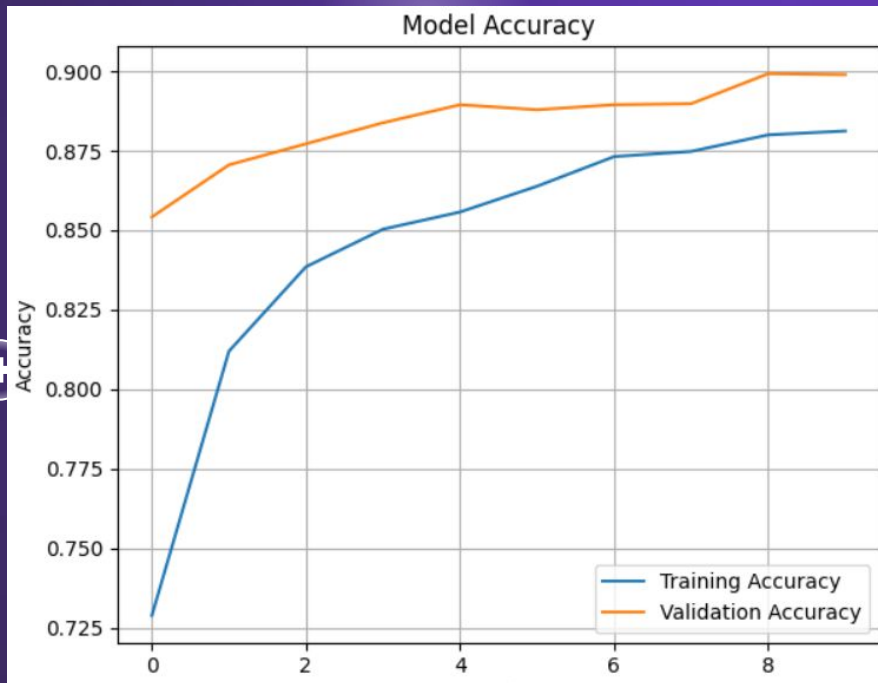
(Before vs After Fine-Tuning)

Metric	Before	After
Training Accuracy	88.12%	93.07%
Validation Accuracy	89.89%	92.72%
Test Accuracy	90.3%	93.36%

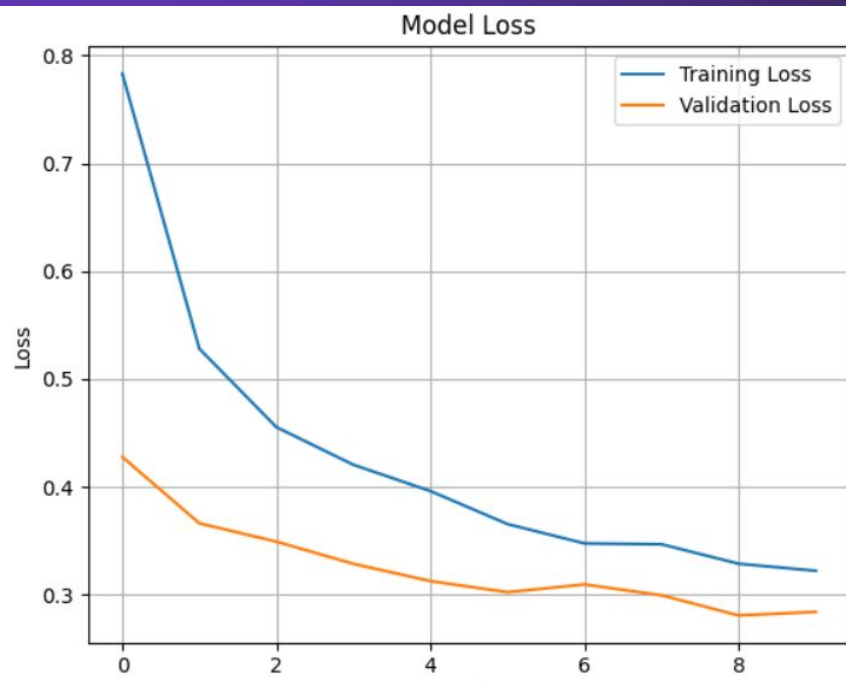




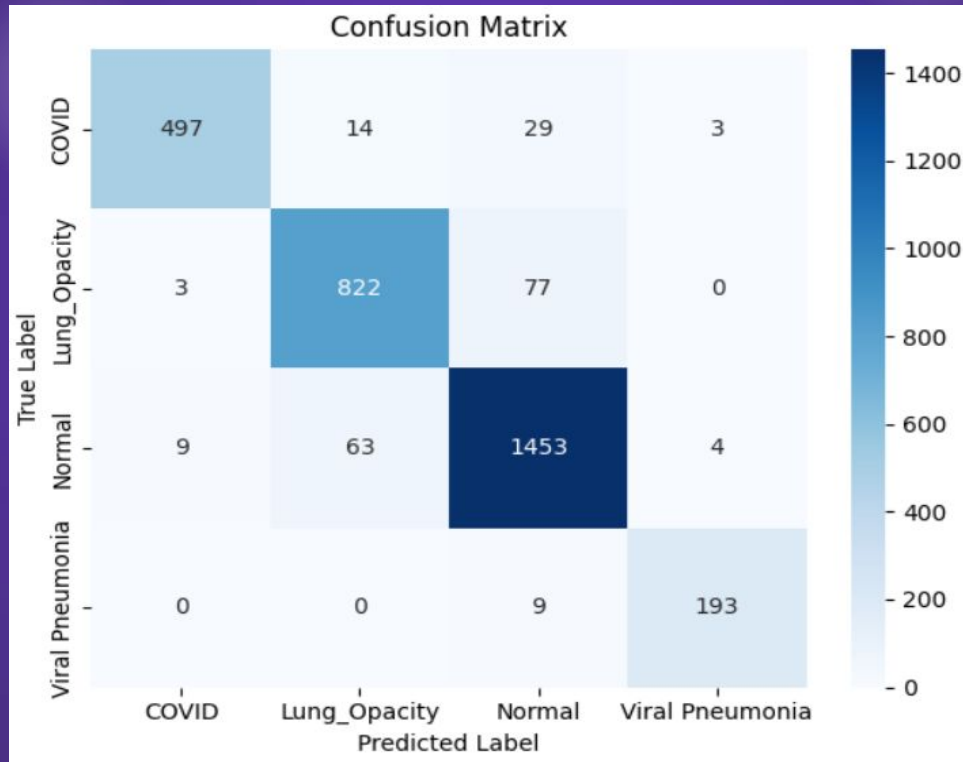
ACCURACY GRAPH

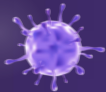


LOSS GRAPH



CONFUSION MATRIX

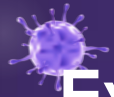




Comparison Table

ACCURACY	CNN 1	CNN 2	RESNET before Fine Tuning	RESNET After Fine Tuning
Training Acc	80.99	80.997	88.12	93.07
Testing Acc	86.24	88.53	90.3	93.36
Validation Acc	85.85	85.85	89.89	92.72





Future Scope : Explainable AI using Grad-CAM for X-ray Classification

What is Grad-CAM?

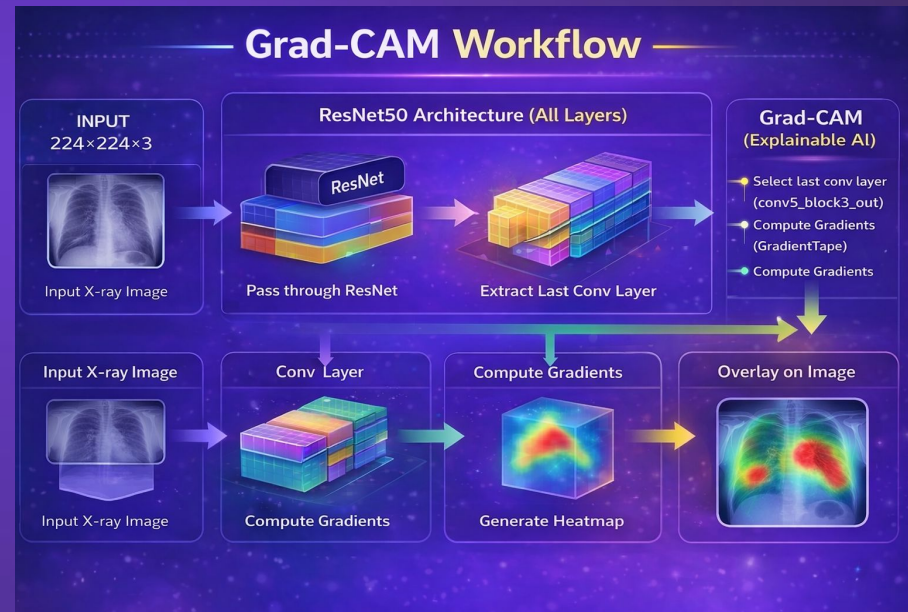
- Gradient-weighted Class Activation Mapping
- Generates **visual explanations**
- Highlights **important regions** in input images

How Grad-CAM Works

- Uses:
 - Feature maps (CNN layers)
 - Gradients of predicted class
- Produces:
 - Heatmap showing decision regions



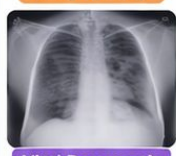
Workflow Diagram



Explainable AI for Medical Image Classification

(CNN with ResNet50 Architecture + Grad-CAM)

Input X-Ray Classes



ResNet50 - All Layers (Detailed)

input_layer (224×224×3)
 conv1_pad (ZeroPadding2D)
 conv1_conv (Conv2D: 7×7, 64)
 conv1_bn (BatchNormalization)
 conv1_relu (ReLU)
 pool1_pad (ZeroPadding2D)
 pool1_pool (MaxPooling2D: 3×3)

conv2_block1_1_conv
 conv2_block1_1_bn
 conv2_block1_1_relu
 conv2_block1_2_conv
 conv2_block1_2_bn
 conv2_block1_2_relu
 conv2_block1_0_conv
 conv2_block1_0_bn
 conv2_block1_0_relu
 conv2_block1_3_bn
 conv2_block1_3_relu
 conv2_block1_1_out

conv2_x (3 blocks)
 conv2_block2_1_conv
 ...
 conv2_block3_out

conv3_x (4 blocks)
 conv3_block1_1_conv
 ...
 conv3_block4_out

conv4_x (6 blocks)
 conv4_block1_1_conv
 ...
 conv4_block6_out

conv5_x (3 blocks)
 conv5_block1_1_conv
 ...
 conv5_block3_out

global_average_pooling2d
 batch_normalization
 dense (FC)
 dropout (0.5)
 (Softmax: 4 classes)

- COVID
- Normal
- Lung Opacity
- Viral Pneumonia

CNN Concept (Feature Extraction)



Grad-CAM (Explainable AI)



How Grad-CAM Works:

- 1 Forward pass → get prediction
- 2 Compute gradients of predicted class
- 3 Weight feature maps (last conv layer: conv5_block3_out)
- 4 Generate heatmap (important regions)
- 5 Overlay heatmap on original X-ray

Why ResNet?

- Skip Connections (Add)
- Solves Vanishing Gradient
- Enables 50+ Deep Layers

Benefits of Explainable AI

- Transparency
- Trust in Predictions
- Medical Decision Support

Key Highlights

- All Layers Shown
- Residual Blocks (conv2_x~conv5_x)
- Grad-CAM for Interpretation

Output Classes

- COVID
- Normal
- Lung Opacity
- Viral Pneumonia

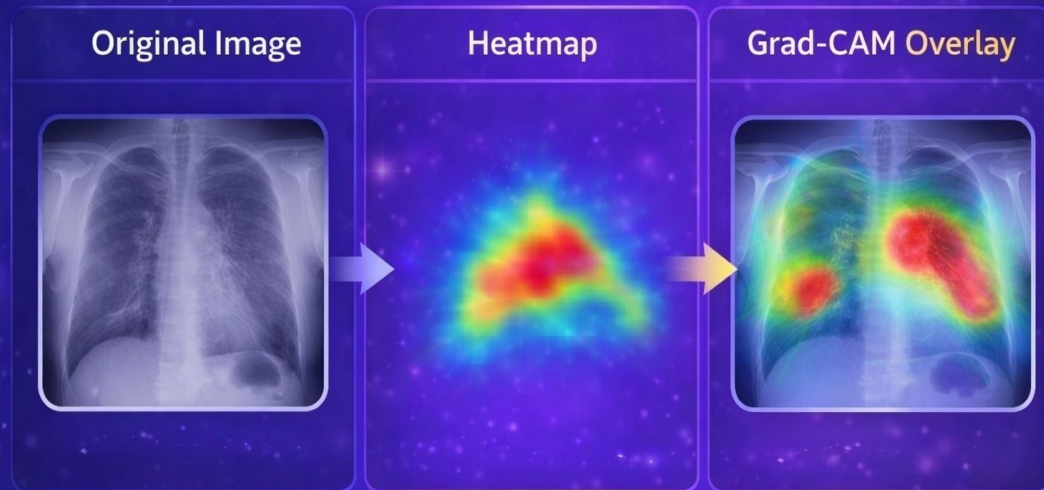
Key Steps in Implementation

- Gradient tracking using GradientTape
- Compute gradients of predicted class
- Global average pooling
- Weighted feature map combination
- Normalize heatmap

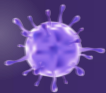
Interpretation :

- **Red** shows highly influential regions (Higher Opacity Region)
- **Yellow** shows moderate importance (Mod Opacity Region)
- **Blue** indicates areas the model largely ignores. (Gaseous Part of Lungs)

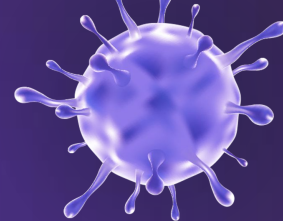
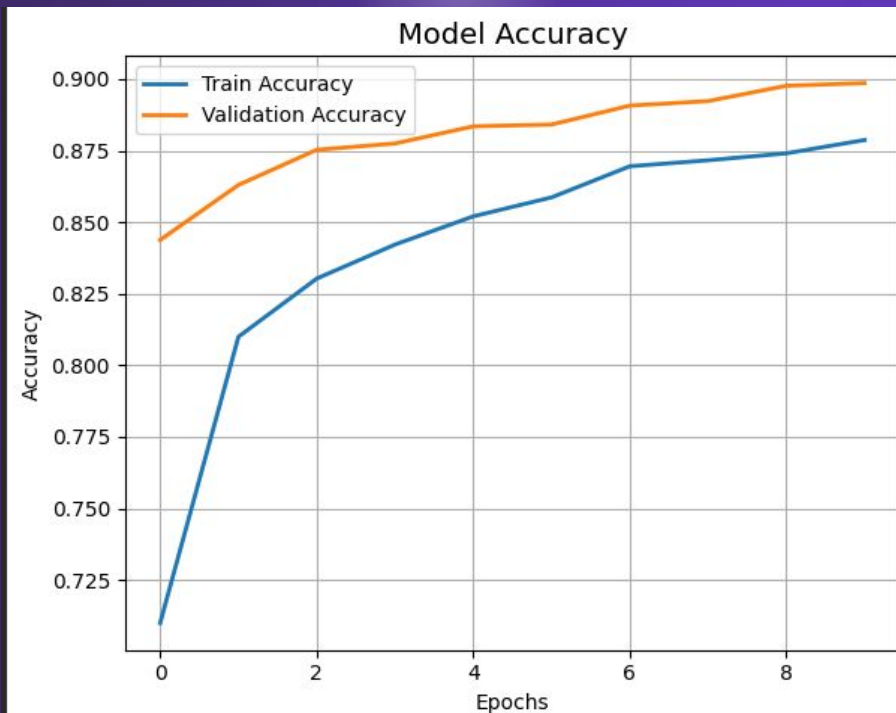
Visualization Output



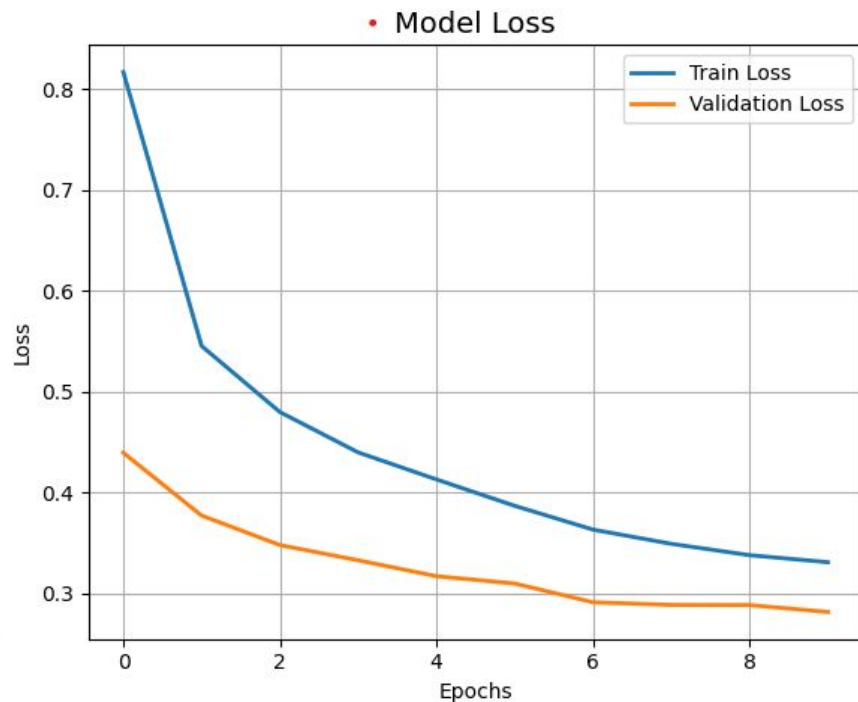
👉 Shows model focus areas

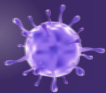


ACCURACY GRAPH

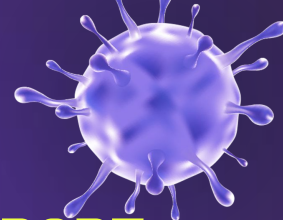
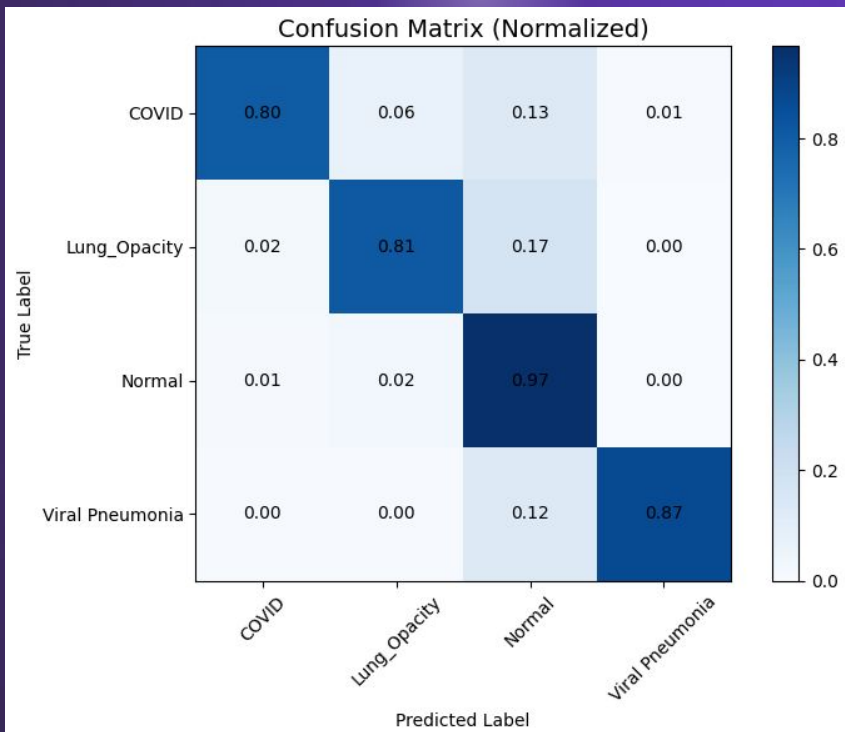


LOSS GRAPH





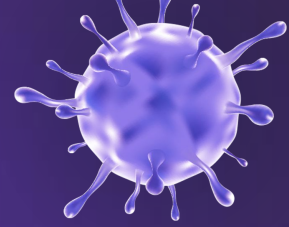
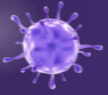
CONFUSION MATRIX



CLASSIFICATION REPORT :

Classification Report:

	precision	recall	f1-score	support
COVID	0.94	0.80	0.87	543
Lung_Opacity	0.91	0.81	0.86	902
Normal	0.86	0.97	0.91	1529
Viral Pneumonia	0.97	0.87	0.92	202
accuracy			0.89	3176
macro avg	0.92	0.86	0.89	3176
weighted avg	0.89	0.89	0.89	3176

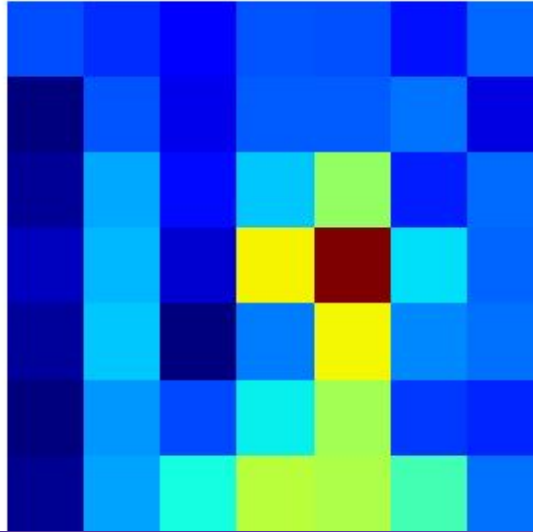


OUTPUT IMG Graded HeatMap :

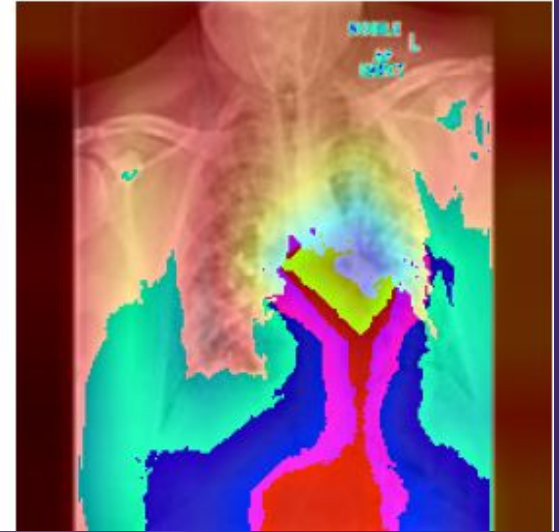
Original

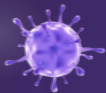


Heatmap



Grad-CAM





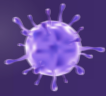
Challenges

- CNN: Overfitting, limited feature learning
- ResNet: High complexity, black-box model
- Grad-CAM: Low resolution, possible misinterpretation

Solutions

- Data augmentation, dropout, normalization
- Transfer learning & fine-tuning
- Use Grad-CAM + expert validation

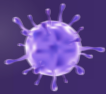




Conclusion

1. Deep Learning Models are effective for covid 19 classification using x ray images
2. CNN Provides good baseline performance.
3. Pre-trained Model (Resnet) improve accuracy using transfer learning .
4. Extended model into an Explainable AI system.
5. Grad-CAM helps improve modal interoperability.
6. Helps identify important regions in X-ray images.





References

1. [Detection of COVID-19, lung opacity, and viral pneumonia via X-ray using machine learning and deep learning - ScienceDirect](#)
2. [COVID-19 Detection Using Deep Learning Algorithm on Chest X-ray Images - PMC](#)
3. [COVID-19 detection from chest X-ray images using transfer learning | Scientific Reports](#)
4. [Detecting COVID-19 From Chest X-Ray Images using CNN - GeeksforGeeks](#)
5. [COVID-19 Radiography Dataset](#)
6. [COVID-19 Dataset](#)
7. [COVID-19 in India](#)





Thank You !!